

SIMULATION-BASED SAFETY SCREENING OF REINFORCEMENT LEARNING ENABLED DRIVING CONTROLLERS USING LOW-FIDELITY MODELS

Suvrat Jain^a, Karrnidh Arora^a, Dev Sharma^a, Jason A. Tanuwijaya^a, Aryan Mukherjee^a, Batuhan Akbas^b
and M. İlhan Akbaş^c

^a Data Science, SP Jain School of Global Management, Sydney, Australia

^b Robert College, Istanbul, Turkey

^c Electrical Eng. & Computer Science, Embry-Riddle Aeronautical Univ., Daytona Beach, FL, USA

ABSTRACT

Simulation results for reinforcement learning–based autonomous driving often fail to translate into reliable real-world behavior, indicating a gap in how early safety is evaluated rather than how controllers are trained. Existing simulation studies predominantly emphasize reward accumulation and task completion, leaving the structure and sensitivity of unsafe behavior insufficiently examined. This paper introduces a simulation-based safety screening approach that reframes low-fidelity simulation as a tool for structured perturbation analysis rather than performance benchmarking. Using CARLoS, a modular open-source low-fidelity driving simulator, the study systematically perturbs road geometry, sensor models and reward parameters to make safety-critical vulnerabilities explicit. The results illustrate how structured perturbations reveal stability limits and unsafe behavioural transitions that may not be apparent under nominal evaluation conditions. This work positions low-fidelity simulation as an effective early-stage screening tool within multi-stage modelling and simulation pipelines and supports principled down-selection of controllers and scenarios for subsequent higher-fidelity simulation.

Keywords: Reinforcement Learning, Autonomous Driving, Low-Fidelity Simulation, Safety Validation

1 INTRODUCTION

Simulation-based development of Reinforcement Learning (RL) controllers has shown significant promise in autonomous driving, where agents learn control policies through interaction with simulated environments rather than relying on explicit rule-based programming [1]. Progress in deep RL has led to successful demonstrations of lane keeping, obstacle avoidance, and complex maneuvering in platforms such as CARLA [2], ANTI-CARLA [3], CARLoS [4], CARLOS [5], Scenic [6], and Waymax [7]. Although these advances illustrate the potential of RL-driven control, most evaluation pipelines rely heavily on reward-based indicators such as accumulated return, task success, and episode duration. These metrics offer only limited visibility into how an agent behaves under uncertainty, how it responds to small disturbances, and how its policy might fail when exposed to conditions that differ from its training distribution. Prior studies show that RL agents often degrade sharply when tested in unseen environments or perturbed scenarios, which reveals the need for broader safety-centered validation methods [2], [8].

A growing body of research emphasizes that reliable autonomous driving requires explicit consideration of robustness, stability, and safety margins rather than simple reward maximization [9]. High-fidelity simulators such as CARLA provide rich visuals, complex physics, and interactive traffic. However, they are often too computationally expensive for large-scale perturbation sweeps, sensitivity analyses, and repeated stress-testing. At early stages of policy development, researchers need environments that support broad scenario variation, rapid iteration, and systematic manipulation of road geometry, sensor characteristics, and reward structure [10]. Low-fidelity simulation therefore plays an important methodological role by enabling fast exploration of behavioural boundaries before more costly testing begins [11]. In this context, the CARLoS simulation framework [4] provides a configurable low-fidelity

environment that is specifically suited for safety-oriented evaluation. CARLoS supports rapid modification of lane width, curvature, obstacle spacing, sensor noise, reward functions, and initial-state offsets. The environment also enables the construction of failure-mode taxonomies and performance-boundary analyses that reveal the conditions under which an RL agent transitions from stable behavior to unsafe operation.

The objective of this paper is to establish a structured methodology for early-stage safety validation using low-fidelity simulation. Rather than positioning CARLoS as a benchmark for algorithmic comparison, we examine how its design enables systematic stress-testing of RL agents under controlled perturbations. The framework supports high-throughput experiments involving initial-state variation, sensor noise injection, environmental distortions, and changes in reward shaping. These experiments illustrate the types of vulnerabilities that can be exposed only when evaluation goes beyond standard training distributions and reward metrics. For example, even small noise or variations in perception can lead to loss of lane stability or unexpected collision events [9]. By applying the framework to multiple representative RL algorithms, we demonstrate the types of safety insights that low-fidelity simulation can reveal. The main contributions of this paper, framed from a modelling and simulation perspective, are as follows:

- A low-fidelity simulation configuration for high-throughput safety screening of learning-enabled controllers, enabling systematic exploration of behavioural limits at low computational cost.
- A perturbation-based simulation experiment design for revealing safety brittleness and robustness properties under controlled variations in environment, sensing, and reward specification.
- A set of quantitative safety metrics suitable for early-stage screening gates, supporting relative comparison and down-selection prior to higher-fidelity simulation or physical testing.

The rest of the paper is organized as follows. Related Work is discussed in Section II. The methodology is given in Section III, and the evaluation is given in Section IV. Finally, the paper concludes in Section V.

2 RELATED WORK

Validation of autonomous driving agents must extend beyond standard training conditions to include controlled perturbations in the form of systematic stress tests [12], [13]. Among the most established simulation environments, CARLA provides a high-fidelity, photorealistic city driving simulator capable of modelling diverse weather, traffic, and pedestrian scenarios. However, its stress-testing capabilities remain limited to specific perturbations such as unseen maps, dynamic obstacles, or adverse weather. Experimental evidence shows that CARLA-trained agents can experience significant degradation in unfamiliar environments, revealing the inadequacy of standard benchmarks for holistic robustness testing. To address this, ANTI-CARLA introduces adversarial and edge-case scenarios, such as erratic pedestrians and sudden obstacles to expose hidden vulnerabilities in RL agents. However, ANTI-CARLA still depends heavily on CARLA, reducing its generalizability to lower-fidelity simulators. Scenic adopts a probabilistic approach to scenario generation by enabling formal specification of diverse driving scenarios with controlled variability in agent behavior and initial states. Its ability to automatically generate thousands of scenario variants strengthens statistical robustness, but it still relies on external simulators such as CARLA and lacks integrated robustness metrics.

To address modularity and scalability limitations, the CARLoS was introduced [4]. CARLoS enables systematic manipulation of road layouts, sensor configurations, and reward definitions for controlled perturbation testing. Its low-fidelity structure prioritizes computational efficiency, enabling large-scale parallel experiments involving sensor noise variations, reward structure changes, and initial-state diversity. This architecture transforms robustness evaluation from a slow, sequential process into a scalable empirical pipeline, accelerating failure-mode discovery.

Low-fidelity simulators such as CARLoS play an increasingly important role in early-stage safety evaluation. Prior studies using CARLA and large-scale simulators like Waymax show that simplified environments still reveal safety-relevant behavioural patterns [14]. Indicators such as lane deviation, off-

road frequency, and collision rate remain meaningful regardless of visual fidelity. CARLoS therefore acts as an early-stage safety filter capable of identifying destabilizing behaviours before high-fidelity testing. This aligns with Gulino et al. [7], who argue that simulation should summarize behavioural variability rather than perfectly mimic reality. Through controlled perturbations, repeated trials and standardized metrics, CARLoS provides efficient pre-deployment safety screening.

Despite progress, key research gaps remain. Cross-simulator transferability, how robustness from low-fidelity translates to high-fidelity simulators, remain understudied. Additionally, systematic comparisons of RL algorithms under standardized stress tests are limited, leaving algorithm-specific robustness characteristics insufficiently explored.

3 METHODOLOGY

This section describes the developed simulation-based safety-validation methodology. The primary objective is to establish a systematic, computationally efficient, and reproducible process for analyzing how RL agents behave under a wide range of controlled perturbations. The methodology is designed to uncover brittle behaviors, characterize stability boundaries, and expose safety-critical failure modes that traditional reward-centric evaluations typically overlook. The pipeline integrates environment specification, agent training, perturbation generation, metric computation, and failure-mode interpretation into a unified framework suitable for early-stage assessment (See Figure 1).

3.1 Simulation Environment

The simulation environment is built on CARLoS, a lightweight open-source simulator designed for configurable testing of autonomy algorithms. A custom Gymnasium-compatible wrapper, CarlosGymEnv, was developed on top of CARLoS to expose it as a standard RL environment compatible with training libraries such as Stable-Baselines3. The developed simulation environment and experiments are publicly available [15].

The environment defines a two-dimensional lane-following task with parametric control over lane width, curvature, and obstacle count, enabling systematic sweeps without modifying any other aspect of the simulation. These parameters can be modified continuously, allowing researchers to build both nominal operating conditions and progressively more challenging safety-relevant scenarios. Obstacles are placed procedurally along the lane at longitudinal positions restricted to $[0.15, 0.85]$ of track length to prevent collisions at spawn, with lateral offsets and radii drawn randomly from discrete sets.

Agents observe the environment through twelve range sensors distributed over a full 360-degree arc with a maximum detection range of 120 units, producing a twelve-dimensional observation vector. The action space is two-dimensional and continuous, with a normalized throttle mapped to a maximum acceleration of 20 mph and a normalized steering command mapped to $\pm\pi/4$ radians. The reward function combines four independently weighted terms: a lane-keeping incentive, a forward-motion incentive, a collision penalty,

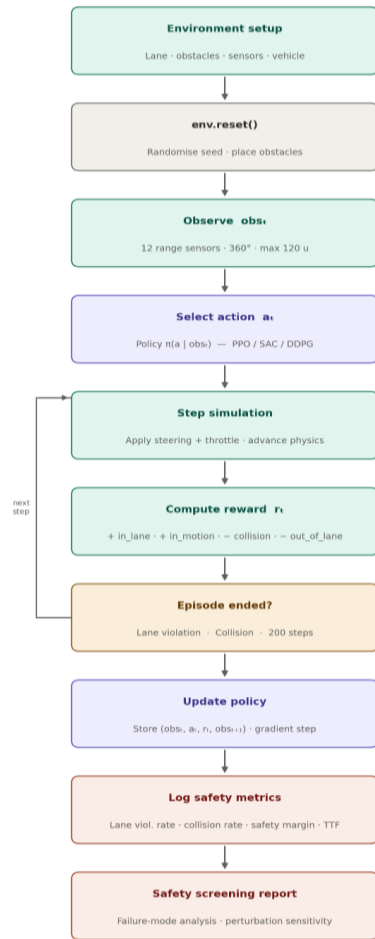


Figure 1. CarlosGymEnv simulation workflow.

and an out-of-lane penalty. An episode ends under one of three conditions: a lane boundary violation, a collision with an obstacle, or successful completion of 200 timesteps. The first two represent safety-critical failures; the third represents sustained safe operation within the evaluation horizon.

The simulator also incorporates disturbances that affect both initial and ongoing agent states. These include initial position offsets, heading errors, sensor noise injections, random curvature perturbations, and abrupt obstacle placement. These disturbances approximate real-world uncertainties and enable evaluation of the agent’s ability to recover from deviations, maintain stability, and avoid escalating into unsafe states.

Together, these elements create a simulation environment optimized for robust, high-throughput safety testing. It serves as a methodological complement to high-fidelity simulators such as CARLA, in which computational cost and visual rendering constraints limit the scale of perturbation-driven experimentation.

3.2 Agent Training Protocol

Agents are trained using Stable Baselines3 (SB3), a standardized deep RL library that provides consistent, reproducible implementations across on-policy and off-policy paradigms. The training protocol follows an identical three-stage curriculum regardless of the algorithm used: Stage 1 (80,000 steps) covers wider lanes (12–14 ft) with low obstacle density (4–6 obstacles); Stage 2 (100,000 steps) introduces tighter geometry (11–12 ft, 6–8 obstacles); Stage 3 (120,000 steps) covers the full perturbation range (10–14 ft, 4–10 obstacles), totaling 300,000 timesteps. At every episode reset, lane width, obstacle count, and reward weights are sampled randomly from the current stage pool across three reward configurations, preventing overfitting to any single condition. Policy weights are preserved across stage transitions; for off-policy algorithms, the replay buffer is reset at each transition so the agent re-explores harder conditions rather than drawing on stale experience from earlier stages.

The general reward function combines four weighted terms computed at each timestep:

$$r_t = w_{\text{in_lane}} \cdot \mathbb{1}[\text{in_lane}] + w_{\text{in_motion}} \cdot \mathbb{1}[\text{in_motion}] + w_{\text{collision}} \cdot \mathbb{1}[\text{collision}] + w_{\text{out_lane}} \cdot \mathbb{1}[\text{out_lane}]$$

where $\mathbb{1}[\cdot]$ is an indicator that is 1 when the condition is true and 0 otherwise.

Config 1 (Balanced) uses {"in lane": +1.0, "in motion": +0.5, "collision": -10.0, "out lane": -5.0}, moderately rewarding lane-keeping and motion while penalizing unsafe events. Config 2 (Motion-encouraged) uses {"in lane": +1.0, "in motion": +1.0, "collision": -5.0, "out lane": -2.0}, emphasizing movement with lighter penalties. Config 3 (Safety-aggressive) uses {"in lane": +2.0, "in motion": +0.3, "collision": -50.0, "out lane": -20.0}, strongly prioritizing safety with heavy penalties for violations.

3.3 Safety Screening Metrics and Decision Criteria

Safety assessment is conducted using six episode-level metrics computed across 100 independent episodes per condition per seed. Lane Violation Rate (LVR) measures the percentage of episodes ending due to a lane boundary crossing, and Collision Rate measures the percentage ending due to obstacle contact. Success Rate captures the percentage of episodes in which the agent completes all 200 timesteps without a safety-critical failure. Time to Failure (TTF) reports the mean number of steps elapsed before a terminal event, computed separately for lane violations and collisions to distinguish the two failure modes. Safety Margin records the mean lateral clearance from lane boundaries across all pre-failure timesteps, with a critical zone defined as clearance below 0.152 m. Control Smoothness quantifies mean step-to-step variation in steering and acceleration signals, serving as an indicator of oscillatory or reactive control behavior that may precede failure. Thus, CARLoS enables fine-grained temporal analysis of how instability emerges, which is essential for characterizing failure progression.

Each condition is further evaluated against four threshold regimes: strict (LVR $\leq$ 20%, collision $\leq$ 5%, TTF $\geq$ 60 steps), standard (LVR $\leq$ 50%, collision $\leq$ 15%, TTF $\geq$ 30 steps), operational (LVR $\leq$ 75%, collision $\leq$ 20%, TTF $\geq$ 20 steps), and lenient (LVR $\leq$ 80%, collision $\leq$ 30%, TTF $\geq$ 10 steps). These regimes span

Tier-1 and Tier-2 safety bands. A condition is classified as "Robust Pass" if it satisfies all four regimes, "Robust Fail" if it satisfies none, and "Marginal" if the verdict changes across regimes. Failed episodes are further categorized into three failure modes: Type I Boundary Drift (gradual lane departure), Type II Abrupt Collision (sudden obstacle contact), and Type III Oscillatory Instability (high-frequency control fluctuation leading to either failure type). This taxonomy enables structured identification of the mechanism of failure rather than simply its occurrence.

3.4 Perturbation-Based Stress Testing

After training, agents are evaluated under 16 perturbation conditions formed by a factorial combination of four lane widths (10, 11, 12, and 14 ft) and four obstacle counts (4, 6, 8, and 10), with the trained policy held fixed throughout. The baseline condition, 12 ft lane width and 6 obstacles, is selected as the nominal reference point, representing a mid-range configuration within the evaluation grid. All other conditions vary lane width and obstacle count relative to this reference, allowing behavioral changes to be interpreted as responses to geometric and density perturbations rather than as absolute performance measures. Each condition is evaluated over 100 independent episodes per seed across three seeds, giving 300 episodes per condition and 4,800 episodes in total. The per-condition metrics are then averaged across the three seeds, with the standard deviation reported alongside the mean to capture seed-to-seed variability. Constant random seeds (42, 68, and 101) are used across all conditions to ensure reproducibility and to attribute observed behavioral differences to the perturbations rather than to stochastic variation in episode initialization. This makes it possible to analyze the way the controller reacts to changes in operating conditions.

3.5 Model Credibility and Use

The CarlosGymEnv framework supports relative comparison and safety analysis of learned policies under controlled perturbations, not real-world performance prediction or safety certification. The framework is intended as an early-stage safety screening tool, serving as a low-cost filter prior to higher-fidelity simulation or physical testing.

4 EVALUATION

Three RL algorithms are selected as representative policy-learning architectures: Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), and Soft Actor-Critic (SAC). Each algorithm embodies distinct control-learning principles and therefore serves as a useful case study for evaluating how the safety-validation methodology generalizes across algorithmic paradigms. SB3 defaults serve as the baseline configuration, and cross-validation is performed to identify optimal values for key hyperparameters, including learning rate, gradient steps, network architecture, and buffer size. All major deviations from SB3 defaults are documented in Table 1.

Table 1: Key hyperparameter settings. Values differing from the SB3 default are shown in the Used column.

Agent	Hyperparameter	SB3 Default	Used
PPO	Learning rate	3×10^{-4} (constant)	$3 \times 10^{-4} \rightarrow 5 \times 10^{-5}$ (linear)
	Gradient steps	N/A (on-policy)	N/A (on-policy)
	Entropy	0.0	0.01
SAC	Learning rate	3×10^{-4}	3×10^{-4}
	Learning starts	100	500
	Gradient steps	1	4
DDPG	Learning rate	1×10^{-3}	1×10^{-4}
	Learning starts	100	500
	Gradient steps	1	4

4.1 Training Performance

Training details of all chosen RL agents are graphically presented in Figure 2.

PPO steadily improves at every step, with longer and more rewarding episodes and no abrupt declines during transitions. By Stage 2, the explained variance increases to about 0.5, and by Stage 3, it stabilizes, signifying critic convergence. This is consistent with learning rate decay, which aids in stabilizing and consolidating the policy as difficulty rises.

SAC shows a surge-and-settle pattern, with rewards repeatedly spiking and correcting across stages. The surges are most prominent in Stage 1 where easier geometry allows broad entropy-driven exploration, and become shallower through Stage 3 as harder conditions constrain the action space. Despite this, stable actor (negative) and bounded critic losses indicate consistent learning. DDPG shows no discernible upward trend even as the curriculum progresses. Large, erratic swings in actor and critic losses, particularly at stage transitions, indicate instability. Without entropy regularization, the policy fails to learn beyond early-stage behavior.



Figure 2 Training diagnostics for PPO, SAC and DDPG.

4.2 Testing Performance

1. PPO Testing

PPO is dominated by boundary drift, with high lane violations (69–85%) and Type I failures across all conditions. Collisions increase with obstacle density (up to ~30%), while success rates stay below 6% and TTF remains low (17–24 steps). Wider lanes improve safety margins slightly, but instability persists, with higher jerk at mid-density. Overall, all four 4-obstacle conditions are Robust FAIL across every threshold regime, the remaining 12 conditions are Marginal passing only under the Lenient Tier-2 regime, and no condition achieves a Robust PASS. All detailed metrics are provided in Tables 2, 3, and 4.

Table 2: PPO outcome metrics across all 16 perturbation conditions.

Lane Width (ft)	Obs.	LVR % (mean $\pm$ SD)	LV Step (mean $\pm$ SD)	Col % (mean $\pm$ SD)	Col Step (mean $\pm$ SD)	Succ % (mean $\pm$ SD)	TTF (steps) (mean $\pm$ SD)
10 ft	4	83.67 $\pm$ 10.41	15.05 $\pm$ 7.53	14.00 $\pm$ 11.79	60.34 $\pm$ 15.53	2.33 $\pm$ 1.53	21.78 $\pm$ 2.19
10 ft	6	75.67 $\pm$ 5.86	13.90 $\pm$ 6.12	21.00 $\pm$ 3.46	55.22 $\pm$ 22.79	3.33 $\pm$ 2.52	22.46 $\pm$ 8.49
10 ft	8	69.33 $\pm$ 3.21	9.11 $\pm$ 0.90	29.00 $\pm$ 2.65	35.87 $\pm$ 4.64	1.67 $\pm$ 2.08	17.10 $\pm$ 1.44
10 ft	10	74.67 $\pm$ 6.11	10.21 $\pm$ 1.13	24.67 $\pm$ 6.66	34.16 $\pm$ 24.29	0.67 $\pm$ 0.58	17.22 $\pm$ 7.54
11 ft	4	83.00 $\pm$ 10.58	15.91 $\pm$ 7.76	13.67 $\pm$ 12.22	44.03 $\pm$ 35.92	3.33 $\pm$ 2.31	22.13 $\pm$ 1.20
11 ft	6	76.00 $\pm$ 6.00	16.33 $\pm$ 9.16	20.33 $\pm$ 5.51	48.50 $\pm$ 15.10	3.67 $\pm$ 2.52	22.91 $\pm$ 7.31
11 ft	8	68.67 $\pm$ 3.06	10.31 $\pm$ 0.28	29.67 $\pm$ 2.31	36.52 $\pm$ 3.58	1.67 $\pm$ 1.15	18.28 $\pm$ 1.66

Lane Width (ft)	Obs.	LVR % (mean $\pm$ SD)	LV Step (mean $\pm$ SD)	Col % (mean $\pm$ SD)	Col Step (mean $\pm$ SD)	Succ % (mean $\pm$ SD)	TTF (steps) (mean $\pm$ SD)
11 ft	10	74.00 $\pm$ 8.00	10.18 $\pm$ 2.33	25.33 $\pm$ 8.50	35.15 $\pm$ 24.97	0.67 $\pm$ 0.58	18.03 $\pm$ 7.01
12 ft	4	82.33 $\pm$ 10.02	16.49 $\pm$ 7.22	14.33 $\pm$ 11.68	44.42 $\pm$ 34.79	3.33 $\pm$ 2.31	22.69 $\pm$ 1.19
12 ft	6	78.33 $\pm$ 7.57	18.11 $\pm$ 11.99	15.67 $\pm$ 5.86	35.22 $\pm$ 8.62	6.00 $\pm$ 2.65	21.52 $\pm$ 8.92
12 ft	8	69.00 $\pm$ 1.73	12.54 $\pm$ 1.78	29.00 $\pm$ 2.00	37.73 $\pm$ 5.66	2.00 $\pm$ 1.00	20.07 $\pm$ 1.85
12 ft	10	76.33 $\pm$ 7.64	12.88 $\pm$ 2.69	23.00 $\pm$ 8.19	38.84 $\pm$ 24.61	0.67 $\pm$ 0.58	20.12 $\pm$ 7.52
14 ft	4	85.00 $\pm$ 9.17	21.13 $\pm$ 5.80	11.00 $\pm$ 10.58	28.42 $\pm$ 18.82	4.00 $\pm$ 2.00	23.71 $\pm$ 1.04
14 ft	6	79.67 $\pm$ 5.03	21.31 $\pm$ 11.02	14.33 $\pm$ 2.89	39.56 $\pm$ 14.60	6.00 $\pm$ 2.65	24.18 $\pm$ 8.46
14 ft	8	70.67 $\pm$ 4.16	15.71 $\pm$ 1.87	26.67 $\pm$ 4.73	39.51 $\pm$ 4.59	2.67 $\pm$ 2.08	22.13 $\pm$ 1.68
14 ft	10	75.67 $\pm$ 6.03	15.32 $\pm$ 1.74	22.67 $\pm$ 7.09	38.08 $\pm$ 23.45	1.67 $\pm$ 1.15	21.58 $\pm$ 7.57

Table 3: PPO safety margin, control behaviour, and dominant failure mode across all 16 conditions.

Lane Width (ft)	Obs.	Safety Margin (m) (mean $\pm$ SD)	Min Safety Margin (m) (mean $\pm$ SD)	Critical Zone % (mean $\pm$ SD)	Control Smoothness (mean $\pm$ SD)	Oscillation Frequency (mean $\pm$ SD)	Jerk (mean $\pm$ SD)	Dominant Failure Type
10 ft	4	2.09 $\pm$ 0.11	1.05 $\pm$ 0.06	0.04 $\pm$ 0.04	0.3855 $\pm$	0.5530 $\pm$	0.2246 $\pm$	TYPE I
10 ft	6	1.90 $\pm$ 0.09	1.00 $\pm$ 0.16	0.03 $\pm$ 0.03	0.3230 $\pm$	0.5232 $\pm$	0.1997 $\pm$	TYPE I
10 ft	8	1.99 $\pm$ 0.24	1.13 $\pm$ 0.09	0.02 $\pm$ 0.04	0.3483 $\pm$	0.5435 $\pm$	0.3014 $\pm$	TYPE I
10 ft	10	2.10 $\pm$ 0.14	1.15 $\pm$ 0.16	0.03 $\pm$ 0.03	0.3613 $\pm$	0.5263 $\pm$	0.2254 $\pm$	TYPE I
11 ft	4	2.28 $\pm$ 0.12	1.10 $\pm$ 0.07	0.03 $\pm$ 0.05	0.3854 $\pm$	0.5496 $\pm$	0.2116 $\pm$	TYPE I
11 ft	6	2.09 $\pm$ 0.13	1.08 $\pm$ 0.13	0.03 $\pm$ 0.03	0.3200 $\pm$	0.5107 $\pm$	0.1917 $\pm$	TYPE I
11 ft	8	2.14 $\pm$ 0.26	1.25 $\pm$ 0.11	0.00 $\pm$ 0.00	0.3420 $\pm$	0.5352 $\pm$	0.3181 $\pm$	TYPE I
11 ft	10	2.30 $\pm$ 0.13	1.19 $\pm$ 0.14	0.15 $\pm$ 0.17	0.3576 $\pm$	0.5272 $\pm$	0.1907 $\pm$	TYPE I
12 ft	4	2.49 $\pm$ 0.14	1.19 $\pm$ 0.08	0.06 $\pm$ 0.03	0.3712 $\pm$	0.5337 $\pm$	0.1924 $\pm$	TYPE I
12 ft	6	2.32 $\pm$ 0.07	1.15 $\pm$ 0.22	0.03 $\pm$ 0.03	0.3308 $\pm$	0.5028 $\pm$	0.1941 $\pm$	TYPE I
12 ft	8	2.30 $\pm$ 0.27	1.30 $\pm$ 0.05	0.04 $\pm$ 0.05	0.3475 $\pm$	0.5291 $\pm$	0.3061 $\pm$	TYPE I
12 ft	10	2.47 $\pm$ 0.16	1.35 $\pm$ 0.13	0.04 $\pm$ 0.04	0.3455 $\pm$	0.5117 $\pm$	0.2052 $\pm$	TYPE I
14 ft	4	2.89 $\pm$ 0.16	1.30 $\pm$ 0.16	0.09 $\pm$ 0.06	0.3370 $\pm$	0.5052 $\pm$	0.1701 $\pm$	TYPE I
14 ft	6	2.73 $\pm$ 0.08	1.35 $\pm$ 0.25	0.05 $\pm$ 0.04	0.3143 $\pm$	0.4784 $\pm$	0.1722 $\pm$	TYPE I
14 ft	8	2.66 $\pm$ 0.30	1.45 $\pm$ 0.10	0.06 $\pm$ 0.05	0.3311 $\pm$	0.5131 $\pm$	0.2754 $\pm$	TYPE I
14 ft	10	2.82 $\pm$ 0.21	1.51 $\pm$ 0.14	0.05 $\pm$ 0.05	0.3266 $\pm$	0.4887 $\pm$	0.1913 $\pm$	TYPE I

Table 4: PPO safety envelope gate verdicts and stability classification, all 16 perturbation conditions.

Condition (Lane Width / Obstacles)	Strict (Tier 1) LVR $\leq$ 20 Col $\leq$ 5	Standard (Tier 1) LVR $\leq$ 50 Col $\leq$ 15	Operational (Tier 2) LVR $\leq$ 75 Col $\leq$ 20 TTF $\geq$ 20	Lenient (Tier 2) LVR $\leq$ 80 Col $\leq$ 30	Stability Classification
10 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
10 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
10 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
10 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
11 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
11 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL

Condition (Lane Width / Obstacles)	Strict (Tier 1) LVR $\leq$ 20 Col $\leq$ 5	Standard (Tier 1) LVR $\leq$ 50 Col $\leq$ 15	Operational (Tier 2) LVR $\leq$ 75 Col $\leq$ 20 TTF $\geq$ 20	Lenient (Tier 2) LVR $\leq$ 80 Col $\leq$ 30	Stability Classification
11 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
12 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
12 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
12 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL

2. SAC Testing

SAC shows the worst safety profile, with extremely high lane violations (80–96%), peaking at 95–96% in 4-obstacle cases. This is driven by the policy’s stochastic entropy-regularized behavior producing unreliable lane adherence even under low obstacle pressure. Collisions increase with obstacle density (up to ~19%), success rates stay below 2.33%, and TTF remains low (14–21 steps) with little improvement from wider lanes. High jerk across conditions reflects unstable, oscillatory control. Overall, 15/16 conditions are Robust FAIL, with only one marginal case under lenient criteria. Thus, establishing SAC as the least safe policy under structured evaluation despite its stable training behavior. All detailed metrics are provided in Tables 5, 6, and 7.

Table 5: SAC outcome metrics across all 16 perturbation conditions.

Lane Width (ft)	Obs.	LVR % (mean $\pm$ SD)	LV Step (mean $\pm$ SD)	Col % (mean $\pm$ SD)	Col Step (mean $\pm$ SD)	Succ % (mean $\pm$ SD)	TTF (steps) (mean $\pm$ SD)
10 ft	4	95.33 $\pm$ 0.58	18.00 $\pm$ 4.12	3.33 $\pm$ 0.58	9.47 $\pm$ 5.84	1.33 $\pm$	17.69 $\pm$ 3.92
10 ft	6	85.00 $\pm$ 3.00	13.25 $\pm$ 6.19	13.67 $\pm$ 2.89	45.62 $\pm$ 31.67	1.33 $\pm$	17.80 $\pm$ 6.56
10 ft	8	81.67 $\pm$ 6.03	10.99 $\pm$ 4.55	18.00 $\pm$ 6.00	26.18 $\pm$ 9.57	0.33 $\pm$	14.16 $\pm$ 1.57
10 ft	10	80.33 $\pm$ 4.04	15.00 $\pm$ 6.67	19.33 $\pm$ 4.51	14.18 $\pm$ 5.83	0.33 $\pm$	14.72 $\pm$ 5.52
11 ft	4	95.67 $\pm$ 0.58	19.45 $\pm$ 5.20	3.33 $\pm$ 1.15	11.08 $\pm$ 11.19	1.00 $\pm$	19.18 $\pm$ 4.95
11 ft	6	84.67 $\pm$ 2.89	13.08 $\pm$ 2.36	13.00 $\pm$ 1.00	52.51 $\pm$ 45.44	2.33 $\pm$	18.01 $\pm$ 7.19
11 ft	8	82.33 $\pm$ 6.11	13.35 $\pm$ 5.26	17.33 $\pm$ 6.03	22.75 $\pm$ 12.15	0.33 $\pm$	15.54 $\pm$ 2.10
11 ft	10	81.67 $\pm$ 3.21	16.08 $\pm$ 7.30	18.00 $\pm$ 3.61	13.34 $\pm$ 6.80	0.33 $\pm$	15.40 $\pm$ 5.68
12 ft	4	95.67 $\pm$ 1.15	19.60 $\pm$ 3.37	3.00 $\pm$ 1.00	6.56 $\pm$ 1.78	1.33 $\pm$	19.21 $\pm$ 3.21
12 ft	6	84.33 $\pm$ 3.06	15.27 $\pm$ 3.17	13.33 $\pm$ 2.52	41.88 $\pm$ 28.92	2.33 $\pm$	18.80 $\pm$ 6.37
12 ft	8	80.67 $\pm$ 6.66	14.09 $\pm$ 6.37	18.67 $\pm$ 6.51	20.70 $\pm$ 11.93	0.67 $\pm$	16.00 $\pm$ 2.58
12 ft	10	80.00 $\pm$ 2.65	16.23 $\pm$ 7.60	19.67 $\pm$ 3.06	13.77 $\pm$ 6.84	0.33 $\pm$	15.60 $\pm$ 6.03
14 ft	4	96.00 $\pm$ 1.00	21.75 $\pm$ 4.09	3.00 $\pm$ 1.73	4.70 $\pm$ 1.81	1.00 $\pm$	21.22 $\pm$ 3.85
14 ft	6	85.67 $\pm$ 2.31	15.90 $\pm$ 3.45	12.00 $\pm$ 2.65	37.36 $\pm$ 42.79	2.33 $\pm$	18.68 $\pm$ 5.29
14 ft	8	86.67 $\pm$ 6.66	18.57 $\pm$ 4.59	12.33 $\pm$ 6.51	16.40 $\pm$ 10.50	1.00 $\pm$	18.81 $\pm$ 2.98
14 ft	10	82.67 $\pm$ 0.58	18.51 $\pm$ 5.47	17.00 $\pm$ 1.00	11.71 $\pm$ 2.61	0.33 $\pm$	17.31 $\pm$ 4.40

Table 6: SAC safety margin, control behaviour, and dominant failure mode across all 16 conditions.

Lane Width (ft)	Obs.	Safety Margin (m) (mean $\pm$ SD)	Min Safety Margin (m) (mean $\pm$ SD)	Critical Zone % (mean $\pm$ SD)	Control Smoothness (mean $\pm$ SD)	Oscillation Frequency (mean $\pm$ SD)	Jerk (mean $\pm$ SD)	Dominant Failure Type
10 ft	4	2.07 $\pm$ 0.07	1.00 $\pm$ 0.09	1.31 $\pm$ 0.07	0.1957 $\pm$	0.4832 $\pm$	0.3782 $\pm$	Type I
10 ft	6	1.97 $\pm$ 0.16	1.02 $\pm$ 0.11	2.20 $\pm$ 0.60	0.1701 $\pm$	0.4907 $\pm$	0.3510 $\pm$	Type I
10 ft	8	2.19 $\pm$ 0.38	1.14 $\pm$ 0.10	1.38 $\pm$ 0.64	0.2142 $\pm$	0.5174 $\pm$	0.3815 $\pm$	Type I
10 ft	10	2.16 $\pm$ 0.33	1.14 $\pm$ 0.12	1.17 $\pm$ 0.70	0.2092 $\pm$	0.4884 $\pm$	0.3381 $\pm$	Type I
11 ft	4	2.31 $\pm$ 0.04	1.15 $\pm$ 0.08	1.29 $\pm$ 0.15	0.2079 $\pm$	0.5002 $\pm$	0.4167 $\pm$	Type I
11 ft	6	2.19 $\pm$ 0.20	1.16 $\pm$ 0.13	1.64 $\pm$ 0.40	0.1780 $\pm$	0.4879 $\pm$	0.4101 $\pm$	Type I
11 ft	8	2.41 $\pm$ 0.33	1.27 $\pm$ 0.09	0.92 $\pm$ 0.23	0.2207 $\pm$	0.5165 $\pm$	0.3810 $\pm$	Type I
11 ft	10	2.39 $\pm$ 0.33	1.27 $\pm$ 0.12	1.05 $\pm$ 0.54	0.2195 $\pm$	0.5089 $\pm$	0.4079 $\pm$	Type I
12 ft	4	2.55 $\pm$ 0.12	1.19 $\pm$ 0.12	1.12 $\pm$ 0.20	0.2164 $\pm$	0.5031 $\pm$	0.4052 $\pm$	Type I
12 ft	6	2.42 $\pm$ 0.20	1.26 $\pm$ 0.09	1.73 $\pm$ 0.57	0.1900 $\pm$	0.4984 $\pm$	0.4000 $\pm$	Type I
12 ft	8	2.60 $\pm$ 0.43	1.45 $\pm$ 0.09	1.08 $\pm$ 0.85	0.2256 $\pm$	0.5166 $\pm$	0.3729 $\pm$	Type I
12 ft	10	2.57 $\pm$ 0.30	1.40 $\pm$ 0.14	1.07 $\pm$ 0.78	0.2350 $\pm$	0.5253 $\pm$	0.4056 $\pm$	Type I
14 ft	4	2.93 $\pm$ 0.11	1.33 $\pm$ 0.12	1.58 $\pm$ 0.18	0.2481 $\pm$	0.5358 $\pm$	0.4179 $\pm$	Type I
14 ft	6	2.81 $\pm$ 0.15	1.48 $\pm$ 0.20	2.14 $\pm$ 0.32	0.2218 $\pm$	0.5365 $\pm$	0.4218 $\pm$	Type I
14 ft	8	2.90 $\pm$ 0.52	1.43 $\pm$ 0.10	1.50 $\pm$ 0.68	0.2554 $\pm$	0.5359 $\pm$	0.3665 $\pm$	Type I
14 ft	10	2.94 $\pm$ 0.25	1.58 $\pm$ 0.06	1.45 $\pm$ 0.66	0.2623 $\pm$	0.5508 $\pm$	0.4036 $\pm$	Type I

Table 7: SAC safety envelope gate verdicts and stability classification, all 16 perturbation conditions.

Condition (Lane Width / Obstacles)	Strict (Tier 1) LVR $\leq$ 20 Col $\leq$ 5	Standard (Tier 1) LVR $\leq$ 50 Col $\leq$ 15	Operational (Tier 2) LVR $\leq$ 75 Col $\leq$ 20 TTF $\geq$ 20	Lenient (Tier 2) LVR $\leq$ 80 Col $\leq$ 30	Stability Classification
10 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
10 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
10 ft / 8 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
10 ft / 10 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 8 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 10 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 8 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 8 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 10 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL

3. DDPG Testing

DDPG shows high lane violations (75–92%), peaking in 4-obstacle cases, with collisions rising to ~23% at higher densities. Success rates are negligible and TTF remains low (14–25 steps). It has the highest jerk,

indicating abrupt, poorly calibrated control. All conditions are Type I failures, with 6/16 Robust FAIL and the rest only marginal under lenient criteria. All detailed metrics are provided in Tables 8, 9, and 10.

Table 8: DDPG outcome metrics across all 16 perturbation conditions.

Lane Width (ft)	Obs.	LVR % (mean $\pm$ SD)	LV Step (mean $\pm$ SD)	Col % (mean $\pm$ SD)	Col Step (mean $\pm$ SD)	Succ % (mean $\pm$ SD)	TTF (steps) (mean $\pm$ SD)
10 ft	4	89.00 $\pm$ 2.00	17.48 $\pm$ 1.94	7.67 $\pm$ 3.06	33.07 $\pm$ 6.96	3.33 $\pm$	18.83 $\pm$ 2.17
10 ft	6	79.33 $\pm$ 4.04	10.66 $\pm$ 2.75	16.33 $\pm$ 2.31	42.89 $\pm$ 6.11	4.33 $\pm$	16.06 $\pm$ 1.84
10 ft	8	76.67 $\pm$ 6.03	10.46 $\pm$ 0.47	22.67 $\pm$ 5.51	38.59 $\pm$ 7.36	0.67 $\pm$	16.68 $\pm$ 0.77
10 ft	10	75.00 $\pm$ 8.54	8.73 $\pm$ 1.18	23.33 $\pm$ 7.02	30.79 $\pm$ 8.44	1.67 $\pm$	14.21 $\pm$ 3.64
11 ft	4	89.00 $\pm$ 2.65	17.88 $\pm$ 2.89	7.33 $\pm$ 1.53	25.90 $\pm$ 6.98	3.67 $\pm$	18.55 $\pm$ 2.12
11 ft	6	78.67 $\pm$ 5.86	14.09 $\pm$ 2.66	16.00 $\pm$ 6.08	40.18 $\pm$ 7.04	5.33 $\pm$	18.78 $\pm$ 3.95
11 ft	8	79.00 $\pm$ 4.58	13.91 $\pm$ 1.96	20.00 $\pm$ 4.58	39.43 $\pm$ 4.60	1.00 $\pm$	19.05 $\pm$ 2.38
11 ft	10	75.00 $\pm$ 7.55	11.57 $\pm$ 1.73	23.00 $\pm$ 6.24	22.83 $\pm$ 8.88	2.00 $\pm$	14.39 $\pm$ 3.31
12 ft	4	89.33 $\pm$ 2.08	20.05 $\pm$ 1.56	7.00 $\pm$ 2.00	31.81 $\pm$ 11.10	3.67 $\pm$	20.99 $\pm$ 1.61
12 ft	6	82.33 $\pm$ 1.15	18.23 $\pm$ 5.60	14.00 $\pm$ 2.00	35.44 $\pm$ 6.12	3.67 $\pm$	20.74 $\pm$ 5.82
12 ft	8	79.33 $\pm$ 7.02	15.03 $\pm$ 2.17	19.67 $\pm$ 6.51	46.46 $\pm$ 11.40	1.00 $\pm$	21.33 $\pm$ 1.69
12 ft	10	77.67 $\pm$ 8.02	15.33 $\pm$ 2.72	21.67 $\pm$ 7.09	24.78 $\pm$ 10.27	0.67 $\pm$	17.64 $\pm$ 4.35
14 ft	4	91.67 $\pm$ 2.52	25.45 $\pm$ 2.47	5.00 $\pm$ 3.00	15.62 $\pm$ 1.19	3.33 $\pm$	24.87 $\pm$ 2.02
14 ft	6	85.33 $\pm$ 1.53	18.29 $\pm$ 3.09	9.67 $\pm$ 1.53	20.58 $\pm$ 5.10	5.00 $\pm$	18.59 $\pm$ 2.63
14 ft	8	75.67 $\pm$ 5.03	15.42 $\pm$ 1.89	21.67 $\pm$ 3.51	33.02 $\pm$ 3.19	2.67 $\pm$	19.46 $\pm$ 0.34
14 ft	10	77.67 $\pm$ 7.64	14.43 $\pm$ 0.71	21.00 $\pm$ 8.54	33.29 $\pm$ 18.99	1.33 $\pm$	19.45 $\pm$ 5.72

Table 9: DDPG safety margin, control behaviour, and dominant failure mode across all 16 conditions.

Lane Width (ft)	Obs.	Safety Margin (m) (mean $\pm$ SD)	Min Safety Margin (m) (mean $\pm$ SD)	Critical Zone % (mean $\pm$ SD)	Control Smoothness (mean $\pm$ SD)	Oscillation Frequency (mean $\pm$ SD)	Jerk (mean $\pm$ SD)	Dominant Failure Type
10 ft	4	2.09 $\pm$ 0.02	0.96 $\pm$ 0.05	0.76 $\pm$ 0.34	0.2607 $\pm$	0.4868 $\pm$	0.5789 $\pm$	Type I
10 ft	6	1.98 $\pm$ 0.07	0.97 $\pm$ 0.16	1.57 $\pm$ 0.15	0.2393 $\pm$	0.4888 $\pm$	0.5643 $\pm$	Type I
10 ft	8	1.99 $\pm$ 0.22	1.04 $\pm$ 0.11	0.82 $\pm$ 0.60	0.2700 $\pm$	0.4942 $\pm$	0.4628 $\pm$	Type I
10 ft	10	2.10 $\pm$ 0.21	1.05 $\pm$ 0.08	0.84 $\pm$ 0.78	0.3007 $\pm$	0.5146 $\pm$	0.5272 $\pm$	Type I
11 ft	4	2.33 $\pm$ 0.08	1.10 $\pm$ 0.07	1.02 $\pm$ 0.19	0.2667 $\pm$	0.4900 $\pm$	0.5724 $\pm$	Type I
11 ft	6	2.17 $\pm$ 0.13	1.04 $\pm$ 0.08	1.79 $\pm$ 0.59	0.2484 $\pm$	0.4907 $\pm$	0.6352 $\pm$	Type I
11 ft	8	2.18 $\pm$ 0.23	1.13 $\pm$ 0.12	0.92 $\pm$ 0.65	0.2745 $\pm$	0.5088 $\pm$	0.5272 $\pm$	Type I
11 ft	10	2.30 $\pm$ 0.18	1.17 $\pm$ 0.09	0.83 $\pm$ 0.69	0.3072 $\pm$	0.5100 $\pm$	0.5390 $\pm$	Type I
12 ft	4	2.57 $\pm$ 0.08	1.16 $\pm$ 0.08	1.25 $\pm$ 0.31	0.2764 $\pm$	0.5071 $\pm$	0.6052 $\pm$	Type I
12 ft	6	2.43 $\pm$ 0.17	1.16 $\pm$ 0.10	1.45 $\pm$ 0.47	0.2727 $\pm$	0.5168 $\pm$	0.5643 $\pm$	Type I
12 ft	8	2.37 $\pm$ 0.28	1.18 $\pm$ 0.07	1.36 $\pm$ 0.95	0.2874 $\pm$	0.5088 $\pm$	0.5376 $\pm$	Type I
12 ft	10	2.53 $\pm$ 0.24	1.29 $\pm$ 0.17	0.79 $\pm$ 0.60	0.3213 $\pm$	0.5291 $\pm$	0.5490 $\pm$	Type I
14 ft	4	3.11 $\pm$ 0.10	1.26 $\pm$ 0.14	1.47 $\pm$ 0.17	0.3108 $\pm$	0.5382 $\pm$	0.5608 $\pm$	Type I
14 ft	6	2.83 $\pm$ 0.03	1.31 $\pm$ 0.23	2.19 $\pm$ 0.33	0.3062 $\pm$	0.5510 $\pm$	0.5654 $\pm$	Type I
14 ft	8	2.84 $\pm$ 0.22	1.40 $\pm$ 0.03	1.66 $\pm$ 0.87	0.3303 $\pm$	0.5589 $\pm$	0.5216 $\pm$	Type I
14 ft	10	2.76 $\pm$ 0.30	1.45 $\pm$ 0.12	1.70 $\pm$ 0.86	0.3360 $\pm$	0.5575 $\pm$	0.5910 $\pm$	Type I

Table 10: DDPG safety envelope gate verdicts and stability classification, all 16 perturbation conditions.

Condition (Lane Width / Obstacles)	Strict (Tier 1) LVR $\leq$ 20 Col $\leq$ 5 TTF $\geq$ 60	Standard (Tier 1) LVR $\leq$ 50 Col $\leq$ 15 TTF $\geq$ 30	Operational (Tier 2) LVR $\leq$ 75 Col $\leq$ 20 TTF $\geq$ 20	Lenient (Tier 2) LVR $\leq$ 80 Col $\leq$ 30 TTF $\geq$ 10	Stability Classification
10 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
10 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
10 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
10 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
11 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
11 ft / 6 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
11 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
11 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
12 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
12 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
12 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 4 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 6 obstacles	FAIL	FAIL	FAIL	FAIL	ROBUST FAIL
14 ft / 8 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL
14 ft / 10 obstacles	FAIL	FAIL	FAIL	PASS	MARGINAL

4.3 Comparing Survivability Across Models

Across all six conditions, PPO consistently maintains the highest survival probability, with SAC showing the steepest early decay and DDPG occupying an intermediate position. The inter-algorithm separation is widest at 14 ft / 6 obstacles, where PPO retains measurable survival beyond step 100 while SAC and DDPG approach zero. As obstacle density increases, this gap narrows, while in the narrowest lane condition, geometric constraint dominates and all three algorithms converge (See Figure 3).

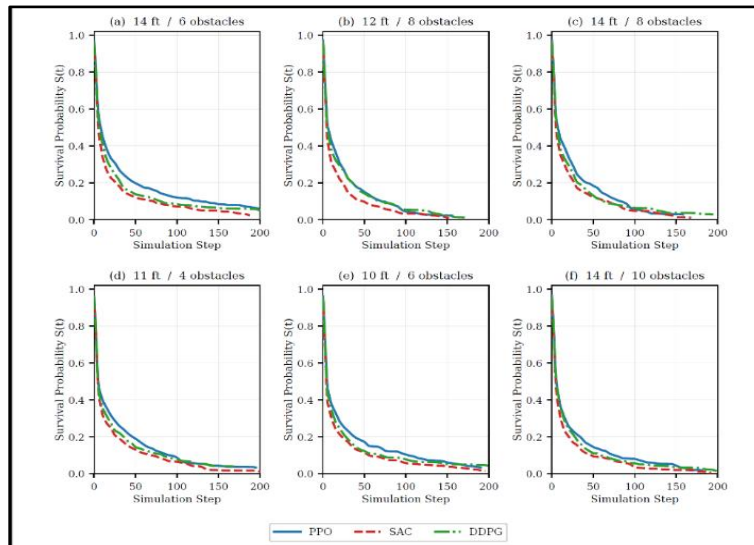


Figure 3: Survival probability comparison across models

5 CONCLUSION

This study presented a systematic process for safety assessment of autonomous driving agents. With sensor model manipulation, reward structure, road geometry and perturbation condition, the framework allowed systematic evaluation of safety critical behaviours not defined by the reward-based measures alone but also by environmental factors. The comparative findings indicated that there were evident variances between the robustness of the used algorithms: PPO shows the best relative profile with 12/16 Marginal conditions, DDPG is intermediate with 6/16 Robust FAIL driven by abrupt deterministic control, and SAC performs worst with 15/16 Robust FAIL despite stable training, caused by its entropy-regularized policy producing unreliable lane adherence. No condition across any algorithm achieves a Robust PASS, confirming that none are deployment-ready and that low-fidelity simulation successfully identifies these deficiencies at low computational cost, fulfilling its role as an early-stage screening filter before higher-fidelity testing.

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AUTHOR BIOGRAPHIES

SUVRAT JAIN is a data science undergraduate at SP Jain School of Global Management, Sydney, Australia, with a keen interest in AI and healthcare integration. His research interest includes developing AI-enabled biomedical devices. His email address is suvrat.bs24dmu007@spjain.org.

KARNIDH ARORA is an undergraduate student at SP Jain School of Global Management, Sydney, Australia, with a focus on machine learning and intelligent systems. Her research interests include adaptive systems and self-learning algorithms; her email address is karnidh.bs24dmu002@spjain.org.

DEV SHARMA is an undergraduate student at SP Jain School of Global Management, Sydney, Australia, with a focus on artificial intelligence and machine learning systems. His research interests include reinforcement learning safety and biometric security. His email address is dev.bs24dmu026@spjain.org.

JASON ADIKA TANUWIJAYA is an undergraduate student at SP Jain School of Global Management, Sydney, Australia, with a deep knowledge in data and AI. His research interests are in machine learning field, algorithm, and artificial intelligence; his email address is jason.bs23dsy031@spjain.org.

ARYAN MUKHERJEE is an undergraduate student at SP Jain School of Global Management, Sydney, Australia, with a focus on machine learning and intelligent systems. His research interests include scalable AI applications and data-driven solutions; his email address is aryan.bs24dmu012@spjain.org.

BATUHAN AKBAS is a student at Robert College in Istanbul, Türkiye. His research interests include topics such as computational thinking, theoretical artificial intelligence concepts, and their real-world applications. His email address is akbbat.27@robcol.k12.tr.

MUSTAFA ILHAN AKBAS is an associate professor at the Electrical Engineering and Computer Science Department, Embry-Riddle Aeronautical University and the director of Flexible and Intelligent Complex System (FICS) Research Group. He received his Ph.D. in Computer Engineering at the University of Central Florida. His research interests include complex systems, wireless communications, modeling & simulation, machine learning and cybersecurity. His email address is akbas@erau.edu.